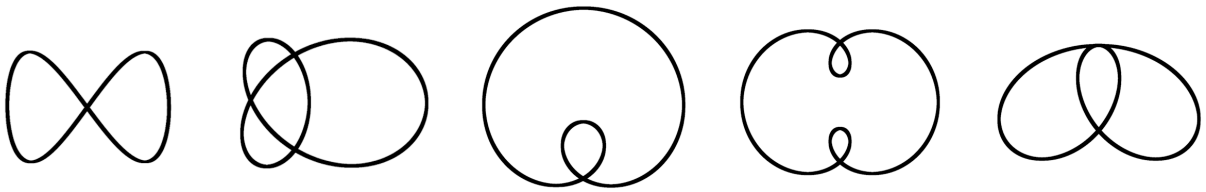
Up From the Hand, Down From the Heart:
Reflections on the Circle Gesture System

Matthew Creighton



1 Introduction

Music is social - what's in an instrument?

Which roles, relationships, ensembles, scenes and communities are hidden in our music-making tools, waiting to be sounded? Which might take work? Which are impossible - precluded by tunings, mappings or interface designs before they're even dreamt of?

If we find that the tools are indeed holding us back - what then? Given the set of musical relationships we *want*, how can we know what instrument to build?



Up From the Hand, Down From the Heart is a research project centred around the building of a new musical instrument - the *Circle Gesture System* (CGS). The outputs are diverse:

- A critique of trends in digital organology from a sociotechnical point of view, and an attempt to build an alternative framework
- A novel mathematical approach to gesture control based around a *symbolic gesture language* called the *Circle Gesture Language* (CGL)
- The new instrument, designed in dialogue with both of the above, and the beginnings of a solo performance practice
- The convening of an ensemble in which to explore the potential of the new instrument and work toward my sociotechnical goals
- The musical output - a 60 minute ensemble improvisation, structured around a text score and a set of collaboratively recorded cassette tape "environments."

The broad spread of these outcomes is part of the point - the project is an effort at integration between several fields i have worked, researched or created in over time. If there's an overall goal, it's an understanding of *communion* - the experience of being present with, and present to, another human - as it relates to musical instruments.

2 The Circle Gesture System



Figure 1: A performer's-eye view of the CGS, showing touchpad, stylus, keyboard, mediator pedal (right) and sustain pedal (left)

The CGS is a digital instrument with no knobs or faders and (almost) no buttons. The performer controls 39 synthesis parameters simultaneously by tracing gestures on a touchscreen with a stylus. A carefully designed language of gestures (the CGL) allows the large number of parameters to be controlled independently even with a two-dimensional input device.

The core idea is *forgetfulness* - leave the synth alone for a while and it returns to its ground state with all parameters zero. No long term state is retained in the form of presets or sample memory, and all mappings between sound and gesture are fixed: the same sequence of actions (if starting from silence) always produces the same sound. This is analogous to an acoustic instrument and in contrast to many electronic instruments.

The left hand plays a small keyboard to choose a single (monophonic) pitch - this action does

not itself produce a sound. Through the touchpad the right hand may play a pure tone at this pitch, add harmonics or inharmonic partials, play a tone at different but related pitches, play chords, or make completely unpitched sounds. The gesture language is *continuous* - blending from any sound to any other is possible by making gradual changes to the drawn gesture. The pitch control in the left hand is 12-tone equal tempered, while the right hand contorl is through pure just intoned ratios. A range of microtonal and timbral effects can be achieved through the tension between these two systems.

The right foot manipulates an expression pedal controller called the *mediator* - intuitively speaking this sets the length of the "short term memory" of the system. Low settings create a tremolo like sound which is very responsive to rapid movement and less sensitive to shape - high settings create a smoother sound in which momentary detail is averaged out and overall shape is more important.

The left foot operates a sustain pedal which works analogously to a piano sustain - if a gesture ends while the sustain pedal is held down, the sound does not die away immediately. In keeping with the principle of forgetfulness, notes held by the sustain pedal do decay slowly - maximum sustain time is around 1.5 minutes.

See DOI (Matthew Creighton 2026) for a video demonstration. Code (including the version used in performance) is publically hosted at (Matt Creighton 2026)

(Choi 1998).

3 A Sociotechnical Framework For Digital Organology

I started working on the conceptual framework underlying the Circle Gesture System after attending AIMC, a conference on AI and music in Brussels. I found the technology interesting and the practitioners kind and thoughtful, but the live performances left me cold. There were a lot of *music agents* (Vincenzo et al. 2025) (Davis et al. 2023) - that is, programs that improvise with humans - and i found myself thinking "well, it would be more fun if there were two people on stage, rather than one and a computer".

That thought wrote itself down as *Don't Eat My Cheeseburger* - we can imagine appreciating a tool that *made* cheeseburgers for us, but there's not many people who want a tool to *eat* it for them too. What need for an improvising agent when improvisation is the fun part?

Over time i wrote several of these aphorisms, collecting them into a manifesto of sorts. Finding fun, memorable (and sometimes polemical) names for things was an important part of the process.

For example, we could think about the *Agency Budget*. In human to human improvisation we know that not everyone can be the centre of attention all the time - in supporting one players agency and expressiveness, others need to show discipline and restraint. So machines that are actively making decisions might be taking up limited expressive "budget" from a human.

I do see that as a problem - which puts me at odds with the move to "biological and relational interfaces" in digital instrument design discussed in (Jensenius and Lyons 2017) - but I'm no luddite. The problem is not technological but *sociotechnical* - to do with the way we *relate to* technology.

A technology with better sociotechnical credentials is *Mario Kart*, the famous Nintendo racing

game. it does well at *Suggesting the Human* - since it's designed to be multiplayer you need to call a friend in order to play it. It prompts a human interaction rather than precluding one.

Thinking about expressivity - good music tools are tools that help a human express whatever theyre feeling that day. They don't mind what the music is "about" - call that *Negotiable Aboutness*. "Composed instruments" and other single use tools possess *Narcisisstic Aboutness* - the tool ends up dictating what the music is about because of its limited palette.



These sociotechnical ideas played into a set of organological concepts related more concretely to design.

The song is in you - Information about what sounds to make and when should reside in the performer. That way, when we see the performer, we're seeing things happen in them, not in the tool. This is what led me to the emphasis on forgetful interfaces, with *no* permanent state residing in the tool..

Fixed Complex Mappings - mappings which change often interfere with muscle memory. So the mappings between sound and gesture should be fixed - but complex enough that they don't need changing between performances.

Comprehensibly Caused - in *A Manifold Interface for Kinesthetic Notation in High-Dimensional Systems* (Choi 1998) Insook Choi discusses a spatial music making system and introduces the idea of a *space proposition* - the idea that in the first few moments of a performance the performer needs to establish vividly the relationship between the sound and their movements in space. This had a big impact on my thinking. I use the more general term *gesture proposition* - the movement to sound relationship should be clear in performance, and in general the sense that the performer is *causing* the sound should be strong after watching the instrument for just a few seconds.

Bring your own binary - Andrew McPherson, in *Notes Aren't Real* (McPherson 2026) discusses how the prevalence of MIDI in digital music pushes us towards design paradigms which are event based, precisely timed and equal tempered, and argues for an "entangled" view of the performer, instrument, and musical setting. I am convinced by the argument - In my own designs I have avoided event based processing which divides time up into discrete "notes". More deeply though I think that coming with some set of binaries and assumtions is inevitable. What I aimed to do in my own work was to present multiple "axes" - note vs noise, equal temperament vs just intonation, responsive and choppy vs sedate and smooth - and allow for rich combinations such that you can "bring your own binary" depending on the day.

4 The Circle Gesture Language

The Circle Gesture Language is the core technology behind the CGS. I'm presenting the mathematical underpinnings at AIMC 2027. In short - it's a *symbolic, continuous, long-sense stateless gesture language* and it's *high- dimensional*.

"Long sense stateless" is the mathematical equivalent of "forgetful" - only things that happened fairly recently affect the output. "Continuous" means that smooth gradations of parameters are possible.

The key innovation in the CGL is that it's *symbolic*, in the traditional semiotic sense. See (Saussure 1959) or (Chandler 2022). Symbolic communication is communication through "signs" - arbitrarily chosen symbols who gain their meaning through opposition with other signs and who's relationship to the referred object is arbitrary. The classic example of a sign is of a word - it's meaning comes not from its sound but its relationship to other words. In developing the CGL i was responding to Rebecca Fiebrink's work on gesture tracing (2009) which continues to be developed at AIMC (O'Donnell and Hartono 2025) . The gesture tracing paradigm asks a user to listen to a pre-recorded track and act out gestures in response. Later the user can make any gesture and an AI system tries to make appropriate sounds based on their responses to the known sound.

This approach can be characterised in Sassure's framework as *mimetic* or *onomatopoeic*- seeking "inherent" meanings in the gesture. I noted that the relationship between gesture and sound on other instruments is fairly arbitrary and symbolic - the finger movements a guitarist makes while playing wouldn't carry inherent meaning if the guitar wasn't there. I theorised that a more densely packed and expressive set of gestures might be possible if we let go of a mimetic approach, and focused simply on designing a well-formed set of gestures. The system of symbols in the CGL is the result of that experimentation.

There are a few gesture parameters that *do* carry deep mimetic meaning in music. returning to the idea of *comprehensible causedness*, we expect larger, faster movements to be louder and perhaps brighter. Alongside the shape tracking code in the CGL I included code to track speed, acceleration and XY position of the stylus. By mapping brightness and volume *mimetically*, through these parameters, we create a sense of conservation of energy and help to make the instrument legible to onlookers. By attaching more abstract timbral parameters to the larger *symbolic* language of gestures, we achieve many dimensions of control, and other "well-formedness" properties such as continuity and parameter independence.

the mediator pedal on the CGS (intuitively speaking) biases the system towards one or other of these systems - fast simple mimetic mappings or slower more expressive symbolic mappings.



it's worth noting that the CGL contains many shapes that are not yet mapped in the CGS. The two diagrams in fig. 2 show the "1-space" (lines, circles, and ellipses, with various sizes and angles - fully mapped in the CGS) and the "2-space" (a more complex set of shapes - only the central block is mapped in the CGS, activating Amplitude Modulation). There are other spaces of equal size to the 2-space that are not mapped at all. Whether full mappings of the CGL with 100+ synthesis parameters are easy to control (or musically useful) remains to be seen)

5 On Building (A Practice)

Building a practice for the new instrument was deeply entwined with building the instrument in the first place. Before I had a good algorithm to recognise CGL shapes i practised drawing them with my fingers on the table. As I built a synthesis engine and a mapping layer and tested out different hardware arrangements, the line between bugs and beloved features became complex. Eventually I ended up with a repertoire both of intended instrumental techniques and unexpected emergent techniques: see DOI (Matthew Creighton 2026)

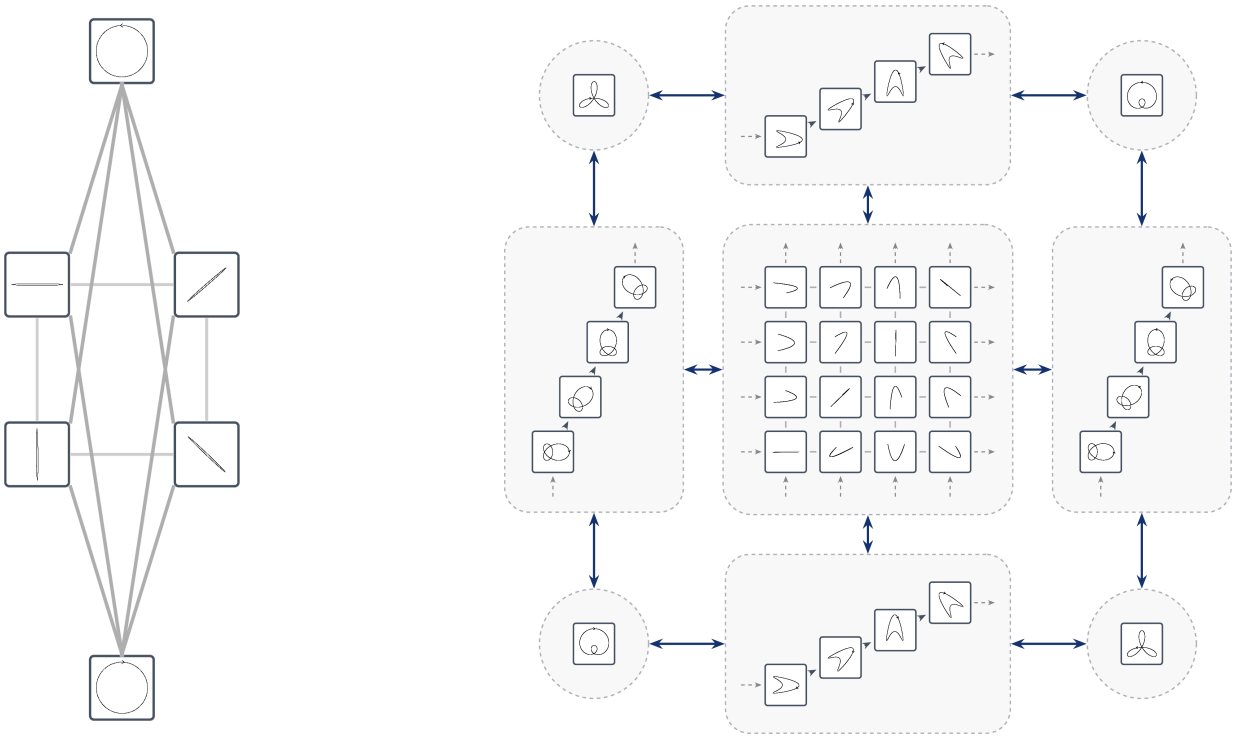


Figure 2: *Left:* CGL 1-space diagram. *Right:* CGL 2-space diagram

A particular place where my theories changed through practice was in the design of the tuning sytem. I did not set out to build a tuning system - I wanted an additive synthesis system where drawing different shapes allowed you add different harmonics to the sound and build any (harmonic) spectrum you wanted. However, three unexpected things happened - first, individual harmonics activated at different times would be heard as different notes: second, those notes formed (some approximation of) a dominant seventh chord (or Lydian Dominant scale), third, those notes were microtonally inflected.

The implied major scale harmony was a major limitation - i remember an early rehearsal asking the ensemble to hold a long minor or suspended chord while I looked for something - anything - that didn't sound like dreamy parallel major chords. So i Introduced some other pitches - this time i chose pitches based on *undertones* - working from Harry Partch 2011 and Ernst Levy 1985 via my favourite book about temperament (Godwin et al. n.d.). The system developed by experimentation - I wanted long straight lines on the touchpad to form different chords depending on directon. fig. 3 shows the system used in the final performance. Markedly microtonal intervals are marked with an asterisk. Figure 4 shows a more realistic view - since the gesture language is continuous, intervals blur smoothly into each other and many chords can be built even without the mediator and sustain.

6 CGS in the Ensemble

I felt sure from the start of the project that I wanted to test the final instrument (and my theoretical ideas) in an ensemble setting. But we can also view the ensemble performances not as a test for the instrument or for me as a performer but as end goals in themselves. After all, my stated aims were sociotechnical - to do with human-human relationships in the presence of

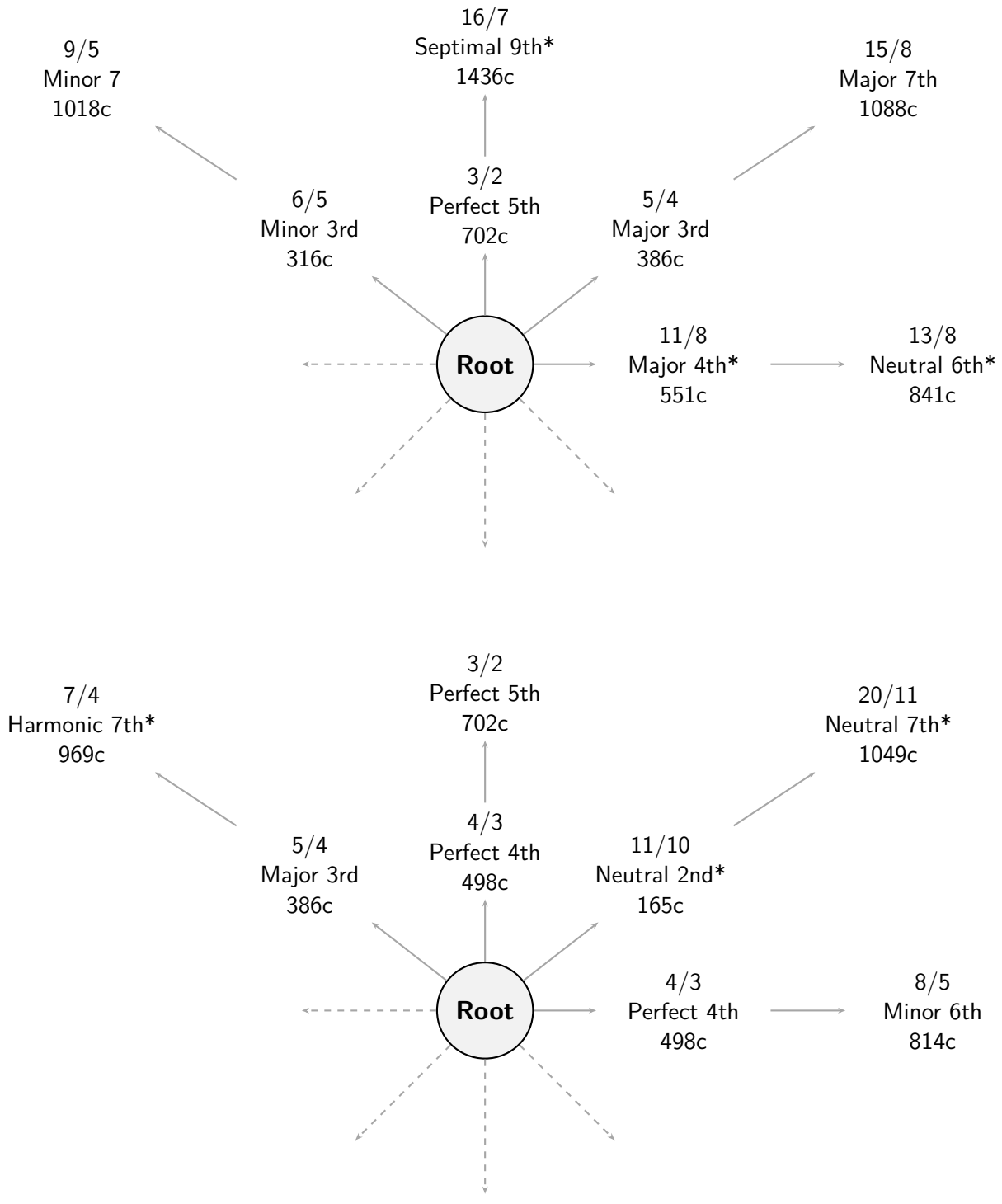


Figure 3: Interval structures for the instrument's tuning systems.

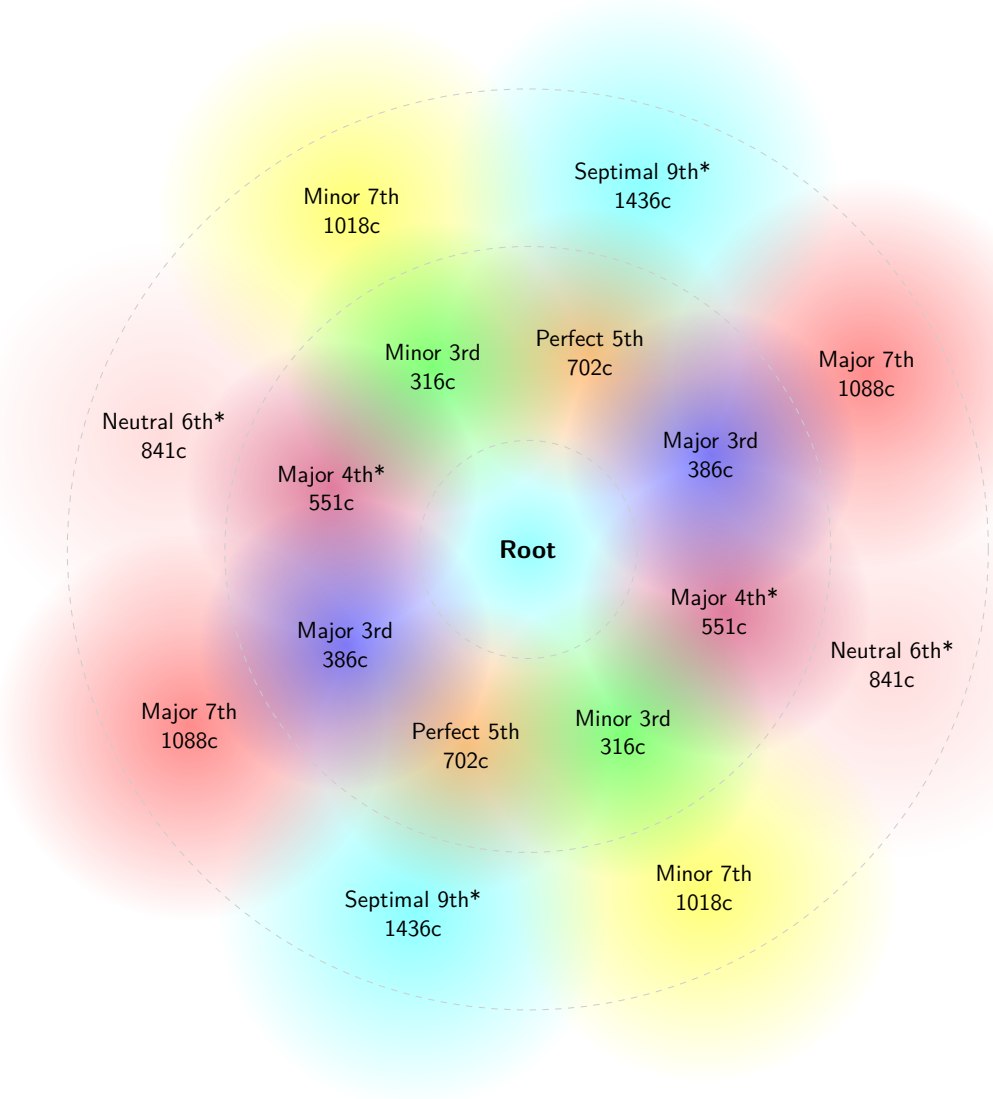


Figure 4: Since the gesture language is continuous, boundaries between pitches are blurry

technology. So by writing music that would be unperformable unless the "right" social setting was present, we can view the performance as an endpoint - as a sign of success.

In thinking about the improvising ensemble in explicitly social terms I was influenced by the concept of a "Game Piece" - most famously enacted by John Zorn (Brackett 2010), also Christian Wolff (Wolff 1998) and - to an extent- La Monte Young in Compositions 1960 (Grimshaw 2011). In these pieces a set of prompts or rules for interaction are described (usually in text form) and the improvisers/performers act a role within these terms.

Game pieces are useful in my case because they allow me to "speak directly to the performers" - an instrument which is stealing the agency budget will get in the way of that. I also have a clear (though hard to express) sense in my own heart of what I want communion between performers to feel like, so i saw this as a chance to reify that too.

See the Supplementary files for a text score.



Each of the text prompts responds to something that, in my view, risks being lost in electronic improvisation at one time or another:

- "Everybody stops at once" is such a reliably exciting mechanism to use in composed performance (especially the guitar music i grew up playing) but it can be hard to coordinate in electronic performance with sequencers, agents and other "uncaused" sounds. for *Theme and Variations* I decided to base the game around moments of sudden silence, Chordrequiring moment to moment interaction between the players to negotiate the timing
- *Chord* is built around similar ideas of human to human communication., asking the players to surrender to a more collective consciousness in performance. I was specifically responding to the *cognitive friction* - that is, needing to think rather than relying on embodied knowledge - that can occur when looking at screens or diving through menus. Such intimate selfless communication becomes very difficult if you also have to think anylitically
- *sounds and silences* is about metre - in a loose abstract sense. I decided to specify that some sort of relationship needs to arise in *when* the performers act and how they make and fill space - which could involve a regular, repeating "clock time" or a more tacit sense of turn taking and respect - what Alfred Schutz refers to as *Duree* (Schütz 1951).
- lastly, *Meeting* responds to the shutting out of the *experience* of improvising when we pass agency over to a machine. The score requires improvisers to come in with one idea and *be changed in performance* by the ideas of other people. This requires both a flexible instrument with negotiable aboutness, and a keen sense of what's going on inside your fellow performers.

7 Tape

The other structuring element in *Up From the Hand, Down From the Heart* is the 4-track tape deck.

Of all the interfaces used by myself and the other performers it presents the most theoretical

problems for me - There's only one sound on (a particular stretch of) the tape, and while the performer can alter this sound, or selectively mute different tracks to tell a story, they're far from *causing* the sound. That sound doesn't respond to who you are or what you're feeling today. Surely this is as bad as a music agent or a composed insgrument from the point of view of this theory?

One way to look at the tape in this project is as a "music thoeretical shadow object" - the harder you work to call something into being (a perfectly forgetful, ultra responsive interface) the more you find yourself in need of its opposite (the primary function of the tape is to *remember* something).

If this sounds a little overblown - it was originally conceived that way explicitly, in an (abortive) earlier iteration of this project. At the time the CGS was more like an effects box that needed an audio input. The combination of a a super flexible tool that couldn't do anything without an audio input, plus a super inflexible tool which could make ANY sound as long as it was prepared ahead of the performance, seemed productive to myself and then-collaborator Amber Perrera.



That particular collaboration was unsucessful. But tape does produce a permanent physical record - a residue. When Sammy Margolis Joined the project we discovered that we'd both owned the same tape machine for a year or two without knowing it, I showed the tapes we had started to prepare to Sammy, and ...we found ourselves recording onto them them together.

Eventually we settled on a model that was respecting of the "caused sound" ideals of the project - the four channels of the tape would create an "environment" rather than a "tracj" and sammy would play the faders and effects quite dynamically in real time. But a durational aspect had to develop - looping on casette tape doesn't produce great results, so to have a sound available on a tape we would fill a whole side of the tape with just that one sound. Some sounds (like synth pads) could be started and left running, but others (like the endless one-chord guitar track on the final tape) had to be recorded in one long take.

I don't know how to theorise the durational element - whether it's opposed to the "forgetful" aesthetic or whether its similar in spirit. In any case, the tape deck (and Sammy expert piloting of it) provided an important counterweight to the abstract logic of the text scores, opened up a space to make the project truly collaborative... and gave me a lot of new theories to try out.

Samples of all the tapes are available in the supplementary files.

8 Bringing It Together

Rehearsal time was short - we had plenty of discussion and planning beforehand, and myself and Sammy had worked on the tapes in his home studio, but we only had two full band meetings before the show. To make sure the others successfully navigated the long score in performance I created a simple hand signals system to guide us from section to section. To make sure **I** navigated successfully, I brought a clock along to the show, set it to 12 noon at the start of the performance, and referred to a circular "setlist" with all the sections marked out as if on a clockface. (conveniently the show is 1 hour i.e one revolution of the minute hand).

To establish a strong gesture proposition at the start of the show I prepared a structured solo improvisation, and a short coda to round things off.

9 Reflections

I will confess to a low feeling right after the performance - I was nervous at the start, and played through my material for improvisation much faster than I had intended. I definitely felt that when I made sounds I didn't like, it threw me in a way it wouldn't usually. I reflect that the "ego risk" was much higher this time: I was playing not just to please an audience or myself but (subconsciously) to convince: to show my first audience that the painstaking development of this instrument was worthwhile.

More prosaically: there was too much in it. I liked the tension between tape and text but we could have taken out Meeting or Theme and Variations and allowed the remaining parts to breathe.

After a night's sleep I felt that the show was sucessful: I learned a lot about the instrument, enjoyed the idiosyncrasies of my collaborators, and we certainly achieved communion at points. I'm really proud of the breadth of different processes I attempted and the cross pollination between them.

Mostly, it feels like the start of a process, not the end of one. Almost every part of this project has gained life of its own: I'm presenting on the mathematics of CGL at AIMC and on its computational aspects at ADC, I'm working towards a second model of the CGS as an entry to the Guthman Musical Instrument Competition (and have started the process to patent the core logic), the ensemble are keen to perform again, and myself and Sammy are planning to explore the the CGS + tape combination in a popular music setting.



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